

Panel MS-AR(1): country intercepts and common growth

1985 through the end of the sample, excluding the US, Germany, Japan, and China

1 Model

Seasonally adjusted real GDP per worker, 2026-09-17 vintage, 1985-Q1 through 2026-Q2. United States, Germany, and Japan are excluded. China is not in this vintage.

$$y_{it} = a_i + g t + z_{it}, \tag{1}$$

$$s_{i,t+1} \sim \Pi(\cdot | s_{it}), \tag{2}$$

$$z_{i,t+1} = (1 - \rho(s_{i,t+1}))\mu(s_{i,t+1}) + \rho(s_{i,t+1}) z_{it} + \sigma(s_{i,t+1}) \varepsilon_{it}. \tag{3}$$

The regime s_{it} is the one that produced z_{it} . ε_{it} is the innovation. For each pair of consecutive dates, g uses the average growth of countries observed on both dates, and those date-pair averages are then averaged with equal weight. a_i is the mean of $y_{it} - gt$ on country i , so $z_{it} = y_{it} - a_i - gt$ has mean zero inside each country. Three regimes. The MS-AR imposes $E[z] = 0$. σ and ρ switch, with $|\rho| < 0.99$. Standard errors are delta-method from a numerical Hessian of the cycle parameters. a_i and g have none.

2 Country intercepts and common growth, 1985 through the end of the sample, ex US, Germany, Japan, China

2026-09-17 SA panel, 1985-Q1 through 2026-Q2, excluding the US, Germany, and Japan. $y_{it} = a_i + gt + z_{it}$, g continuing-country growth, $E[z] = 0$. 71 countries, 7893 observations, 1985-Q1–2026-Q2. Fitted countries: 71. Observations: 7893. Log-likelihood: 21127.41. Continuing-country growth is $g = 0.0183$ (no standard error). Country intercepts a_i have mean -5.6820, minimum -7.7272, and maximum -3.5745 (no standard errors). a_i is the intercept at the first date in the estimation sample. $z_{it} = y_{it} - a_i - gt$ has mean zero in each country and is the cycle in the figure. A country's slope of z is its own growth minus g . The MS-AR is the law of z , with $E[z] = 0$.

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.1049 (0.0234)	-0.0857 (0.0375)	0.0945 (0.0136)
σ	0.0075 (0.0002)	0.0782 (0.0033)	0.0187 (0.0006)
ρ	0.9900 (0.0000)	0.9564 (0.0117)	0.9819 (0.0021)
$\Pi_0.$	0.9462 (0.0060)	0.0106 (0.0036)	0.0432 (0.0070)
$\Pi_1.$	0.0032 (0.0097)	0.6488 (0.0371)	0.3479 (0.0389)
$\Pi_2.$	0.0480 (0.0063)	0.0567 (0.0069)	0.8953 (0.0097)
π	0.4318 (0.0310)	0.0903 (0.0101)	0.4780 (0.0265)

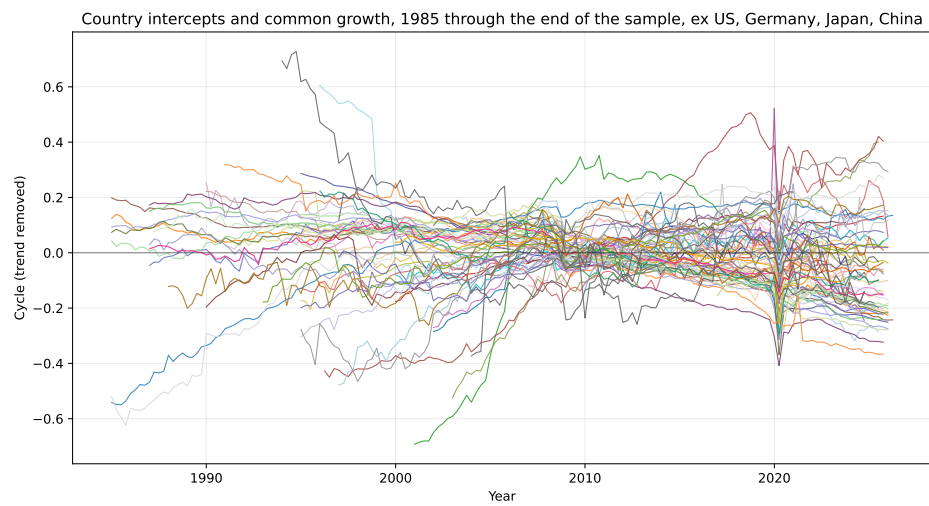


Figure 1: Country cycles $z_{it} = y_{it} - a_i - gt$.